ZJUI 2026 Summer Research Showcase
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Audio-Text Driven 3D Digital Human Synthesis and Emotional Expression

Introduction

Background and Significance

Most digital human systems rely on single-modality driving. Audio-only models merely produce lip-sync faces, while text motion generators fail to output render-compatible 2D hand keypoints. Full-body models lack delicate hand motions, and popular semi-body pipelines demand manual pose inputs, creating high user thresholds.

Research Objectives

1. Bridge HandMDM and EchoMimicV2 to convert 3D motion sequences into 2D pose keypoints.
2. Generate fine-grained 3D hand motions from text, paired with audio for synchronized facial and lip animation.
3. Construct a lightweight modular end-to-end renderer outputting 768 × 768 24fps high-res upper-body videos with simple inputs.
4. Realize smooth, unified multi-emotional coordination of body and facial motions.

Methodology

Model Architecture

1. Text Gesture Module: HandMDM takes text prompts to output 25fps 274D SMPL-H motion features with upper-body and detailed 3D hand motions.
2. Bridge Conversion Module: Adapts motion data via frame interpolation, 6D forward kinematics, weak perspective projection, and MANO-to-DWPose 21-hand keypoint conversion.
3. Rendering Module: EchoMimicV2 fuses reference image, audio features and converted 2D poses to generate digital human videos through diffusion.

Data Sources

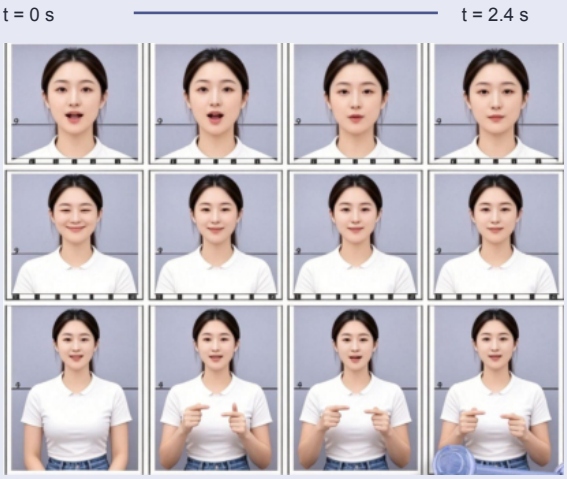
1. BOBSL3DT: 1.31M text-motion pairs to train HandMDM hand gestures.
2. EMTD: Real high-res upper-body videos for identity-consistent rendering.
3. Inputs: Text, WAV audio and one frontal reference image.

Analysis Tools

1. Deep Learning Frameworks: PyTorch, HuggingFace Diffusers
2. Pose Modeling: SMPL-H/MANO, DWPose full-body keypoint detector
3. Geometry Tools: 6D rotation transform, FK, weak perspective projection
4. Visualization Tools: OpenCV, Matplotlib, MoviePy

Results

Core Data Figures



Visual Analysis

Analysis 1

Our system can generate temporally smooth multi-frame upper-body sequences with consistent character identity, synchronized lip movements, matching facial emotions and diverse fine hand gestures under joint text-audio driving. Different emotional states (calm, smiling, speaking) maintain natural coordination between face and hands without jitter, texture distortion or hand deformation artifacts.

Analysis 2

Our model outperforms baseline methods across core evaluation metrics:

1. Visual quality: Lower FID/FVD, higher SSIM and PSNR, indicating clearer, more realistic generated videos.
2. Audio-lip synchronization: Higher Sync-C and lower Sync-D scores, realizing precise alignment between speech content and lip movements.
3. Hand animation: Higher HKC (hand keypoint confidence) and HKV (hand keypoint variance), proving stable and diverse hand gesture generation.
4. Identity retention: Higher CSIM (cosine similarity of facial features), ensuring consistent facial features across all video frames.

Conclusion

Research Significance

This work combines HandMDM and EchoMimicV2 to build a dual-modal digital human system with superior rendering performance; future optimizations will simplify the pipeline and expand capabilities.

Future Work

Future work will integrate the pipeline, support full-body animation, enrich emotional expressions and speed up real-time generation.

References

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